

Project Design Phase

Proposed Solution Template

Date	19 July 2026
Team ID	[Enter Your Team ID]
Project Name	BookEase – Multi-Role Online Bookstore Platform
Maximum Marks	2 Marks

The project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Readers struggle to discover and buy books at fair prices from a single trusted source, small independent sellers lack an affordable way to reach online customers, and platform operators need centralized control to keep a multi-seller marketplace trustworthy.
2.	Idea / Solution description	BookEase is a MERN-stack, multi-role online bookstore. Customers browse, search, wishlist and buy books with a persistent cart and tracked order history. Independent sellers register, get admin-approved, and list/manage their own inventory. Admins oversee sellers, users, books, categories and orders from a single dashboard.
3.	Novelty / Uniqueness	A lightweight multi-seller marketplace model with an approval-gated seller onboarding flow; a client-side cart cache that syncs with the server so the cart never appears to "lose" items across logins; and mandatory reason-tracking on order cancellations that feeds back into seller/admin insight.
4.	Social Impact / Customer Satisfaction	Gives small and independent booksellers/publishers digital reach without needing their own e-commerce infrastructure. Gives readers a wider genre selection, transparent pricing (with GST/discount breakdown), and full order/cancellation transparency.
5.	Business Model (Revenue Model)	Commission on each seller transaction, optional seller subscription/listing fees for premium catalog placement, delivery-charge margin on orders below the free-delivery threshold, and coupon-driven promotional tie-ups with publishers.
6.	Scalability of the Solution	A stateless REST API and MongoDB Atlas allow horizontal scaling of both frontend and backend independently. The role-based architecture (customer/seller/admin) can accommodate new roles — e.g., a delivery-partner role — without redesigning the core system.